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ARCH–LSTM Structural Equivalence with Hybrid Student-t Likelihood Loss for Financial Volatility Forecasting

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Abstract

Accurate conditional volatility forecasting is essential for risk management, asset pricing, and portfolio optimization. Despite the widespread use of GARCH family models and the proliferation of deep learning architectures, a methodological gap persists: neural networks are rarely trained under criteria statistically coherent with the distributional properties of financial returns. This paper makes two contributions. First, we establish formal structural equivalences between classical heteroscedastic models and neural architectures, showing that ARCH(p) is equivalent to a single-layer linear MLP and GARCH(1,1) to a constrained LSTM, with an explicit parameter correspondence. Second, we propose LSTM-SSE-t-Student, a parsimonious LSTM trained with a hybrid loss that combines the sum of squared errors with the Student-t negative log-likelihood, penalizing errors in the tails of the return distribution. The model is evaluated on six daily series spanning three asset classes—Bitcoin, Ethereum, Gold, Oil, the DJIA, and the S&P 500—across diverse regimes, including the COVID-19 period, against a broad set of econometric and deep learning benchmarks. Relative to GARCH(1,1), it significantly improves point forecast accuracy and probabilistic calibration over naive, short-memory, and regime-switching specifications, while matching the strongest GARCH family and deep learning competitors; a sensitivity analysis shows that the likelihood term lowers the QLIKE loss for most series, with a market-dependent optimal weighting. Statistical significance is assessed via Diebold–Mariano tests with a correction for multiple comparisons, and Value-at-Risk and Expected Shortfall backtests confirm adequate tail calibration for the equity and cryptocurrency series. Interpretability is preserved through the GARCH-consistent structure, whose learned gate dynamics are stable across random seeds.

Keywords: volatility forecasting; LSTM; GARCH; ARCH; hybrid loss function; deep learning; financial time series; conditional heteroscedasticity

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1. Introduction

The precise estimation of conditional volatility in financial return series constitutes a cornerstone of modern quantitative finance. Volatility directly determines the computation of risk measures such as Value-at-Risk (VaR) and Expected Shortfall (ES), underpins the pricing of derivative instruments, governs optimal portfolio allocation strategies, and informs regulatory capital requirements under frameworks such as Basel III (Basel Committee on Banking Supervision, 2011) Consequently, the development of models that accurately capture the dynamic structure of conditional variance has been a persistent focus of both academic research and financial practice.

The classical paradigm for volatility modeling rests on the family of Autoregressive Conditional Heteroscedasticity (ARCH) models introduced by Engle (Engle, 1982) and subsequently extended to the Generalized ARCH (GARCH) formulation by Bollerslev (Bollerslev, 1986). These specifications offer parsimony, economic interpretability, and a well-established estimation framework grounded in maximum likelihood inference. The GARCH(1,1) model has demonstrated remarkable empirical performance across a wide range of asset classes, capturing canonical financial stylized facts, including volatility clustering, heavy-tailed return distributions, and mean reversion in variance. Nevertheless, the parametric constraints of ARCH/GARCH models limit their capacity to represent non-linear dynamics, structural breaks, and higher-order dependencies that characterize contemporary financial markets (Bhambu et al., 2025; Brik & Ouakdi, 2024; Jia & Yang, 2021; Siddiraju & Hasan, 2025; Y. Zhang et al., 2020).

Recent work has converged on three benchmark families that we analyze here: convolutional recurrent hybrids such as CNN-LSTM (Choi & Shin, 2022; Vidal & Kristjanpoller, 2020), attention-based and fully attentional sequence models (Chen et al., 2025; Ramos-Pérez et al., 2021; Y. Zhang et al., 2025), and hybrids that retain a parametric variance recursion, including ANN-GJR, Markov-switching Beta-t-EGARCH and GARCH-LSTM designs (Baffour et al., 2019; Di-Giorgi et al., 2025; Kim & Won, 2018; Liao et al., 2020). Against these, parsimonious linear benchmarks such as HAR remain competitive (Corsi, 2008; Vortelinos, 2015), and surveys report gains that vary substantially across markets and architectures (Ge et al., 2022; Qiu et al., 2025).

A further and more fundamental gap concerns the theoretical relationship between econometric and neural models. A growing body of work has paired GARCH-type dynamics with recurrent architectures, but almost always at the level of empirical hybridization or informal analogy rather than formal correspondence. Several studies embed GARCH-generated volatility estimates as inputs to an LSTM, or train a recurrent network to reproduce GARCH-like behavior, and observe that the resulting models perform competitively (García-Medina & Aguayo-Moreno, 2024; Durairaj & Mohan, 2019; Liu & So, 2020; Siddiraju & Hasan, 2025; Tripathy, 2025; C. Zhang et al., 2024).

These contributions establish that recurrent networks can approximate conditional variance dynamics, and some note qualitatively that the LSTM cell state resembles the recursive variance accumulation of GARCH(1,1). What none of them provides, however, is an explicit, parameter-level mapping between the two model classes: the analogy remains descriptive, the correspondence between specific GARCH coefficients and specific network components is left unspecified, and consequently the equivalence cannot be exploited either to constrain the network to a known econometric solution or to interpret its learned dynamics in econometric terms.

Our work precisely closes this gap. Rather than treating the GARCH–LSTM relationship as a motivating intuition, we state it as a pair of formal propositions with explicit parameter correspondences—ARCH(p) to a single-layer linear MLP, and GARCH(1,1) to a constrained LSTM in which the forget gate, input gate, and cell state map to the persistence coefficient, the ARCH coefficient, and the conditional variance, respectively. This formalization is not an end in itself: it is what makes the two central uses of the equivalence possible. First, it allows econometric restrictions to be transferred directly into the network as hyperparameter constraints, placing the classical GARCH solution demonstrably within the LSTM parameter space. Second, it turns the network's gate dynamics into interpretable, econometrically meaningful quantities, enabling the empirical validation of the equivalence through gate inspection reported in Section 4.5. It is this move, from informal resemblance to an exploitable, parameter-level correspondence, that distinguishes the present contribution from the prior hybrids cited above.

This paper addresses both gaps through two principal contributions. First, we provide formal propositions of structural equivalence between ARCH(p) and linear MLP networks, and between GARCH(1,1) and constrained LSTM networks, with explicit parameter mappings that facilitate the transfer of econometric restrictions into neural hyperparameter constraints. Second, building on this theoretical foundation, we introduce LSTM-SSE-t-Student, a parsimonious LSTM architecture trained with a hybrid loss function that combines SSE with the negative log-likelihood under a Student-t distribution, directly mirroring the GARCH maximum likelihood estimation criterion under heavy-tailed innovations.

The proposed model is evaluated on six heterogeneous financial series spanning three asset classes: two cryptocurrencies (BTC-USD, ETH-USD), two major equity indices (DJIA, S&P 500), and two commodities (Gold, Oil), encompassing observation windows that span multiple volatility regimes. The comparison set is deliberately broad, comprising econometric benchmarks (ARCH(1), GARCH(1,1), EGARCH, GJR-GARCH, FIGARCH, MSGARCH, and HAR) alongside machine learning and deep learning benchmarks (SVR-GARCH, NN-GARCH, an MSE-trained LSTM, CNN-LSTM, LSTM-Attention, TCN, and Transformer), so that any improvement can be attributed to the proposed hybrid loss rather than to the relative simplicity of the comparison set. Forecasting accuracy is assessed through both point and probabilistic criteria, and the statistical significance of the differences is evaluated via the Diebold–Mariano test with a correction for multiple comparisons, complemented by Value-at-Risk and Expected Shortfall backtests. The remainder of the paper is organized as follows. Section 2 presents the theoretical framework: Section 2.1 reviews the related literature, Section 2.2 establishes the formal ARCH–neural equivalence, and Section 2.3 describes the proposed model. Section 3 details the experimental setup, including the datasets, the benchmark models, the evaluation metrics, and the computational environment. Section 4 reports the results, covering out-of-sample performance, residual diagnostics, sensitivity to the hybrid loss hyperparameters, gate dynamics analysis, and risk backtesting. Section 5 discusses the findings and their limitations, and Section 6 concludes.

2. Theoretical Framework

2.1. Related Work

Financial volatility, conventionally defined as the standard deviation of returns conditional on available information, reflects the degree of uncertainty in an asset's price evolution. Since volatility is latent and not directly observable, empirical estimation relies either on proxies, typically squared or absolute returns, or on parametric specifications of the conditional second moment. The ARCH family (Bollerslev, 1986; Engle, 1982) provides the dominant framework, formulating conditional variance as a linear function of past squared innovations and, in the GARCH extension, of past conditional variances. Extensions addressing specific empirical regularities include EGARCH (Nelson, 1991), which accommodates the leverage effect through an asymmetric exponential formulation; GJR-GARCH; and fractionally integrated variants (FIGARCH, FIEGARCH) for long-memory processes (Baillie et al., 1996). These models are estimated via maximum likelihood under specified distributional assumptions, with the Student-t and skewed-t distributions offering improved fit relative to the Gaussian in the presence of heavy tails (Fernández & Steel, 1998).

The development of high-frequency data has promoted realized variance (Andersen et al., 2003) as an improved volatility proxy, and Heterogeneous Autoregressive (HAR) models (Corsi, 2008) capture multi-scale persistence in realized volatility through parsimonious multi-lag specifications. While these models are interpretable and well-calibrated under their distributional assumptions, their linear structure limits adaptation to the non-linear volatility dynamics observed in contemporary markets.

The application of deep learning to financial volatility forecasting has substantially expanded over the past decade. LSTM networks (Hochreiter & Schmidhuber, 1997) and Gated Recurrent Units (Cho et al., 2014) have demonstrated strong performance in modelling the sequential dependencies of financial time series. Multiple studies report that LSTM-based models achieve lower out-of-sample forecasting error than GARCH benchmarks, particularly under volatile regimes and for highly non-linear assets such as cryptocurrencies. Hybrid architectures combining econometric and deep learning components, including models that embed GARCH-generated volatility estimates as inputs to LSTM networks, have attracted particular interest, as have Transformer-based architectures and attention mechanisms for long-range dependency modelling.

A consistent finding is that deep learning models achieve superior point accuracy relative to econometric benchmarks, particularly for longer forecast horizons and in markets with structural breaks. However, these improvements are frequently obtained at the cost of interpretability and probabilistic calibration, as MSE-trained networks implicitly assume Gaussian innovations and systematically underfit the tails of the return distribution.

The formal relationship between GARCH-type models and recurrent neural architectures has been explored in several contributions (García-Medina & Aguayo-Moreno, 2024; Siddiraju & Hasan, 2025; Yang et al., 2024). Early work observed that GARCH models can be viewed as special cases of neural networks with specific weight constraints. The "LSTM as GARCH" formulation establishes that the cell state update mechanism of an LSTM replicates the recursive variance accumulation structure of GARCH(1,1). However, these treatments have remained largely informal, without providing the systematic parameter correspondence necessary to exploit the equivalence for constrained training or rigorous interpretability analysis. The present paper closes this gap by providing formal propositions with explicit mathematical derivations, enabling both theoretical analysis and principled model design.

The use of maximum likelihood-based loss functions in neural network training for financial applications has gained increasing attention. Training recurrent networks under Student-t likelihood rather than MSE has been shown to improve tail behavior in residuals and enhance performance on extreme event evaluation metrics. Asymmetric loss functions that differentially penalize under- and over-forecasting of volatility have also been proposed, reflecting the asymmetric economic consequences of risk misestimation. The present study is distinctive in that the choice of t-Student likelihood loss is not motivated empirically alone but grounded in the formal GARCH equivalence established in Section 2.2, which implies that this loss is the theoretically appropriate training criterion for an LSTM designed to replicate GARCH dynamics.

2.2. Theoretical Framework: ARCH–Neural Network Structural Equivalence

ARCH(p) and GARCH(p, q) Specifications. Let r_t denote the log-return of a financial asset at time t, computed as $r_t = ln(P_t/P_{t-1})$, where P_t denotes the closing price. The centered return is $\varepsilon_t = r_t - \mu$, where μ is the conditional mean, assumed constant for simplicity. The ARCH(p) model (Engle, 1982) specifies:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \cdots + \alpha_p \varepsilon_{t-p}^2 \quad (1)$$

with constraints $\omega > 0$, $\alpha_i \geq 0$, and $\Sigma_i \alpha_i < 1$ for covariance stationarity. The GARCH(p, q) extension GARCH (Bollerslev, 1986) augments the conditional variance with q lags of σ^2:

$$\sigma^2{}_t = \omega + \Sigma_i \alpha_i \varepsilon^2{}_{t-i} + \Sigma_j \beta_j \sigma^2{}_{t-j} \quad (2)$$

The GARCH(1,1) special case (the most widely employed variant) reduces to $\sigma^2{}_t = \omega + \alpha\varepsilon^2{}_t{}^{-1} + \beta\sigma^2{}_t{}^{-1}$, with stationarity requiring $\alpha + \beta < 1$. Parameters are estimated by maximising the log-likelihood. Under the Student-t distribution with ν degrees of freedom, the log-likelihood contribution at time t is:

$$\ell_t = ln\Gamma((\nu+1)/2) - ln\Gamma(\nu/2) - \tfrac{1}{2}ln\left[\pi(\nu-2)\sigma^2{}_t\right] - ((\nu+1)/2)ln\left[1+\varepsilon^2{}_t/\left((\nu-2)\sigma^2{}_t\right)\right] \quad (3)$$

MLP and LSTM Architectures. A single hidden layer MLP with input vector $x \in \mathbb{R}^p$ takes the form $f(x;\theta) = W^2 \cdot \varphi(W^1 x + b^1) + b^2$, where W^1, W^2 are weight matrices, b^1, b^2 are biases, and $\varphi(\cdot)$ is an element-wise activation function. With linear activation ($\varphi(x) = x$), the output reduces to a linear projection of the input. The LSTM network (Hochreiter & Schmidhuber, 1997) processes sequential inputs via four coupled gated transformations:

$$\begin{aligned} i_t &= \sigma\left(W^I x_t + U^I h_{t-1} + b^I\right) \quad \text{[input gate]} \\ f_t &= \sigma\left(W^f x_t + U_f h_{t-1} + b^f\right) \quad \text{[forget gate]} \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad \text{[output gate]} \\ g_t &= tanh\left(W^G x_t + U^G h_{t-1} + b^G\right) \quad \text{[cell candidate]} \\ c_t &= f_t \odot c_{t-1} + i_t \odot g_t \quad \text{[cell state]} \\ h_t &= o_t \odot tanh(c_t) \quad \text{[hidden state]} \end{aligned} \quad (4)$$

where $\sigma(\cdot)$ denotes the logistic sigmoid and $\odot$ denotes element-wise multiplication.

Proposition 1. *ARCH(p) ≡ Linear Single-Layer MLP. Consider an MLP with a single hidden layer, linear activation, input vector* $\left[\varepsilon^2{}_{t-1}, \ldots, \varepsilon^2{}_{t-p}\right]$*, and a non-negative output constraint. The network output is algebraically identical to the ARCH(p) conditional variance when the composed weight matrix* $W^2W^1 = \left[\alpha^1, \ldots, \alpha_p\right]$ *and output bias* $b^2 = \omega$.

Proof. With $\varphi(x) = x$, the output collapses to $f(x;\theta) = \left(W^2W^1\right)x + b^2$. Setting $W^2W^1 = \left[\alpha^1, \ldots, \alpha_p\right]$ and $b^2 = \omega$ *yields* $f\left(\varepsilon^2{}_{t-1:\,t-p}\right) = \omega + \Sigma_i \alpha_i \varepsilon^2{}_{t-i}$, precisely the ARCH(p) specification. Non-linear activations extend this, allowing the MLP to capture non-linear volatility dynamics beyond ARCH. □

Proposition 2. *GARCH(1,1) ≡ Constrained LSTM. Consider an LSTM with scalar input* $x_t = \varepsilon^2{}_t$. *Impose: (i) forget gate* $f_t = \beta$ *(constant); (ii) input gate* $i_t = \alpha$ *(constant); (iii) cell candidate* $g_t = \varepsilon^2{}_t$ *; (iv) output gate* $o_t = 1$ *; (v) initial cell state* $c^0 = \omega/(1-\alpha-\beta)$. *Under these constraints, the cell state update becomes:*

$$c_t = \beta \cdot c_{t-1} + \alpha \cdot \varepsilon^2{}_t \quad (5)$$

which is formally equivalent to the GARCH(1,1) recursion $\sigma^2{}_t = \omega + \alpha \cdot \varepsilon^2{}_{t-1} + \beta\sigma^2{}_{t-1}$, with the intercept ω absorbed into the initial condition. The cell state c_t thus serves as the conditional variance $\sigma^2{}_t$; the forget gate coefficient β replicates the GARCH persistence parameter; and the input gate coefficient α replicates the ARCH coefficient. The unrestricted LSTM generalizes this by allowing adaptive gate dynamics.

This equivalence implies that the GARCH(1,1) solution lies within the LSTM parameter space and that training under a loss function equivalent to the GARCH log-likelihood guar-

antees that the landscape includes the classical econometric solution. The full parameter correspondence is summarized in Table 1.

Table 1. Structural parameter correspondence between ARCH/GARCH and neural architectures.

Model	Component	Neural Analogue	Constraint
ARCH(p)	$\varepsilon^2{}_{t-i}$ (squared innovations)	Input vector x	-
ARCH(p)	α_i (ARCH coefficients)	Weight matrix W^2W^1	Non-negative
ARCH(p)	ω (intercept)	Output bias b^2	Positive
GARCH(1,1)	$\sigma^2{}_t$ (cond. variance)	Cell state c_t	-
GARCH(1,1)	α (ARCH coeff.)	Input gate i_t	Constrained to scalar
GARCH(1,1)	β (GARCH coeff.)	Forget gate f_t	Constrained to scalar
GARCH(1,1)	ω (intercept)	Initial cell state c^0	Scaled by $(1-\alpha-\beta)$
MLP/LSTM	Non-linear activations	Extension beyond ARCH/GARCH	None

2.3. Proposed Model: LSTM-SSE-t-Student

Figure 1 illustrates the structural equivalence described in Section 2.2; the proposed architecture consists of a single-layer LSTM with one cell, followed by a dense output layer with softplus activation $[f(x) = ln(1+e^x)]$ to enforce strict positivity of the conditional variance estimate. The input at each time step is a sliding window of squared centered returns $\left[\varepsilon^2{}_{t-1}, \ldots, \varepsilon^2{}_{t-w}\right]$, with window size $w = 1$ for the primary specification, directly analogous to the ARCH(1) lag structure. This minimalist architecture deliberately mirrors the parsimony of GARCH(1,1) while allowing learned gate dynamics to generalize beyond its linear constraints. Only the final hidden state h_t is forwarded to the dense output; a dropout layer, with a rate determined by hyperparameter tuning, is inserted between the LSTM output and the dense layer.

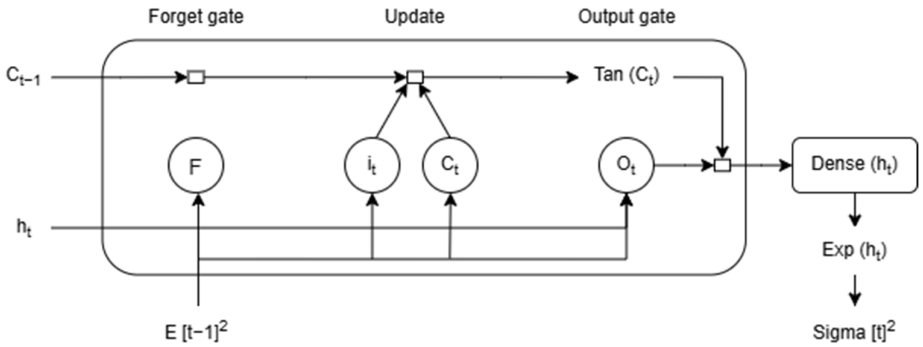


Figure 1. General architecture of the proposed LSTM network. Inputs are squared centered returns; the output is the estimated conditional variance $\hat{\sigma}_t^2$.

The proposed loss function L^{hybrId} combines two complementary criteria:

$$L^{\text{hybrId}} = (1-\lambda)\cdot L_{SSE} + \lambda \cdot L_{t-student} \tag{6}$$

where $\lambda \in [0,1]$ is a mixing parameter tuned during optimisation:

$$L_{SSE} = (1/T)\Sigma_t\left(\hat{\sigma}_t^2 - \varepsilon^2{}_t\right)^2 \tag{7}$$

$$L_{t-student} = (1/T)\Sigma_t ½ln\hat{\sigma}_t^2{}_t + ((\nu+1)/2)ln\left[1+\varepsilon^2{}_t/\left(\nu\hat{\sigma}_t^2\right)\right] \tag{8}$$

Here, $\hat{\sigma}_t^2$ is the network output, ε_t is the centered return, and $\nu > 2$ is the degrees of freedom parameter treated as a tunable hyperparameter. The SSE component penalizes pointwise error and provides gradient stability during early training. The Student-t

negative log-likelihood introduces an asymmetric penalty: for large $|\varepsilon_t|$ relative to $\hat{\sigma}_t^2$ the gradient decays polynomially rather than exponentially as under Gaussian loss, ensuring a sustained gradient signal for extreme events that determine VaR and ES at high confidence levels.

Critically, Equation (8) is formally equivalent to the GARCH log-likelihood criterion under Student-t innovations (Equation (3)), rendering LSTM-SSE-t-Student statistically comparable to its econometric counterparts and guaranteeing that the GARCH solution lies within the optimization landscape. The parameter ν governs tail heaviness: values in the empirically motivated range $\nu \in 3, \ldots, 8$ imply substantial excess kurtosis consistent with observed return distributions.

The model is optimized using the Adam algorithm (Kingma & Ba, 2014) with cosine decay learning rate scheduling. The initial learning rate is drawn from $\left[10^{-2}, 10^{-4}\right]$ and decays to a minimum of 10^{-6}, mitigating oscillations near convergence. Gradient clipping with a threshold of 1.0 prevents gradient explosion under extreme volatility regimes, particularly prevalent in cryptocurrency markets.

Table 2 shows that the hyperparameter optimization via Keras Tuner explores: LSTM units in {16, 32, 64, 128}; degrees of freedom ν in {3, 4, 5, 6, 7, 8}; dropout rate in {0.0, 0.1, 0.2, 0.3}; mixing parameter λ in {0.1, 0.3, 0.5, 0.7, 0.9}; batch size in {32, 64, 128}; and up to 1000 training epochs with early stopping. Temporal cross-validation with five-fold TimeSeriesSplit preserves chronological order and prevents look-ahead bias. All experiments fix random seeds in TensorFlow (V. 2.17.0), NumPy (V. 1.26.4) , and Python (V 3.11.6, with Glorot uniform weight initialization. The search space for λ excludes the boundary values 0 and 1: λ = 0 corresponds to the LSTM-SSE baseline and is evaluated separately, while λ = 1 is excluded because the pure Student-t loss may produce unstable gradients during early training epochs, a known issue that the SSE component is specifically designed to mitigate.

Table 2. Hyperparameter search space for the LSTM-SSE-t-Student model.

Hyperparameter	Range/Values	Selection Criterion
LSTM units	{16, 32, 64, 128}	Minimize validation loss
Degrees of freedom (ν)	{3, 4, 5, 6, 7, 8}	Minimize negative log-likelihood
Dropout rate	{0.0, 0.1, 0.2, 0.3}	Temporal cross-validation
Mixing parameter (λ)	{0.1, 0.3, 0.5, 0.7, 0.9}	Balance SSE and likelihood
Initial learning rate	$\left[10^{-2}, 10^{-4}\right]$	Cosine decay to 10^{-6}
Batch size	{32, 64, 128}	Minimize validation loss
Max epochs	Up to 1000	Early stopping on validation loss
Validation scheme	TimeSeriesSplit, $k = 5$	Prevent look-ahead bias

The output layer applies softplus activation, ensuring $\hat{\sigma}_t^2 > 0$ for all inputs without the zero-gradient regions of ReLU that would cause numerical instabilities in log-likelihood computation. Inner LSTM gates use standard sigmoid (input, forget, output) and hyperbolic tangent (cell candidate) activations per Equation (4). Batch normalisation is applied to the dense layers after the LSTM output to stabilize gradient magnitudes across heteroscedastic financial series.

3. Materials and Method

3.1. Datasets

The empirical evaluation employs six daily financial time series spanning three asset classes and multiple market environments. For the cryptocurrency domain, daily closing prices of Bitcoin (BTC-USD) and Ethereum (ETH-USD) were obtained from Yahoo

Finance; the BTC-USD series began on 28 April 2013 (n = 4304), while ETH-USD began on 7 August 2015 (n = 3155). These markets are characterized by extreme volatility, heavy-tailed return distributions, and pronounced non-stationarity, constituting demanding test cases for conditional variance models. For the equity domain, daily closing prices of the Dow Jones Industrial Average (DJIA) and the Standard & Poor's 500 (S&P 500) were extracted over the period beginning 1 June 2015 ($n \approx 2786$). For the commodity domain, daily prices of gold (Gold) and crude oil (Oil) were obtained over a comparable period beginning 1 June 2015 ($n \approx 2786$), extending the evaluation to markets subject to infrequent but severe price jumps. All six series were updated through 30 June 2026, and this common end date encompasses multiple volatility regimes, including the COVID-19 turbulence of early 2020. See Figure 2. In every series, the chronological 80/20 train–test split preserves temporal ordering, with a validation partition drawn from the end of the training window for early stopping and hyperparameter tuning; the out-of-sample window accordingly ends on 30 June 2026. No shuffling was used to preserve temporal integrity, and all random seeds were fixed to ensure reproducibility.

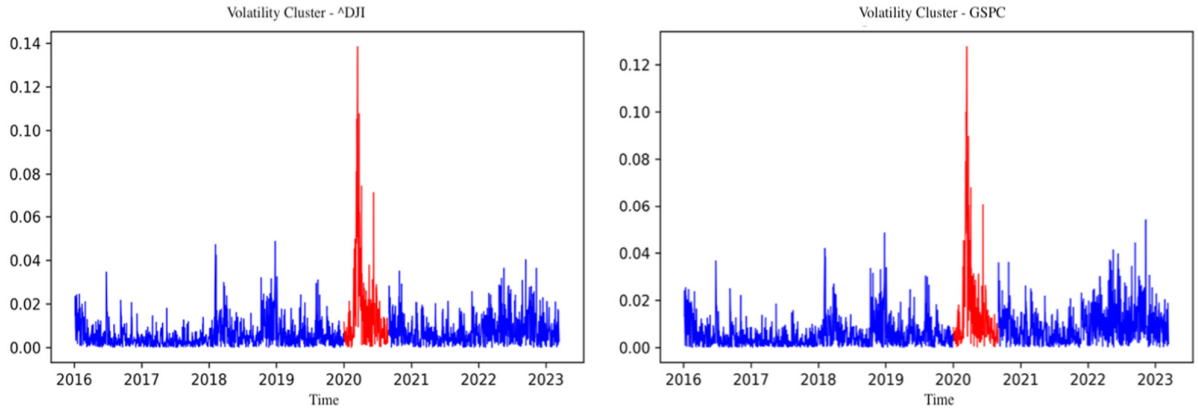


Figure 2. Conditional variance estimated by ARCH(1) for DJIA (**left**) and S&P 500 (**right**). The COVID-19 volatility cluster (January–June 2020) is highlighted in red.

Table 3 reports descriptive statistics for daily log-returns of the six series (BTC-USD, ETH-USD, DJIA, S&P 500, Gold, and Oil) over the sample period. All series exhibited negative skewness, indicating a higher propensity for large negative return shocks than for positive ones—a well-documented stylized fact in financial markets. More critically, excess kurtosis was substantial across all assets (ranging from 9.81 for Gold to 21.94 for DJIA), far exceeding the zero value implied by the normal distribution, confirming pronounced leptokurtosis and fat tails in the unconditional return distributions. The highly significant Ljung-Box Q^2 statistics and ARCH-LM tests (both at the 1% level) rejected the null hypotheses of no serial correlation in squared returns, providing strong evidence of volatility clustering and conditional heteroscedasticity. Taken together, these results justify two core methodological choices adopted in this study: (i) the use of Student-t innovations rather than Gaussian innovations in the conditional variance models, since Gaussian-based likelihoods would systematically understate the probability of extreme events; and (ii) the specification of a hybrid Student-t likelihood loss for the proposed LSTM architecture, which explicitly accounts for fat-tailed conditional distributions rather than relying on a symmetric squared error loss alone. Ignoring this excess kurtosis would bias volatility forecasts and understate tail risk, undermining the reliability of any downstream VaR/ES backtesting. As a visual example, Figure 2 shows in red one of the most volatile periods in the DJI and SP500 series, associated with market instability caused by COVID-19.

Table 3. Descriptive statistics of daily log-returns for the six evaluation series.

Series	N	Mean	Std. Dev.	Skewness	Excess Kurtosis	$LBQ^2(20)$	$LBQ^2(20)$ *p*-Value	ARCH-LM(20)	ARCH-LM(20) *p*-Value
BTC-USD	4304	0.1127	3.5072	−0.7059	11.8567	257.503 ***	0.0000	164.232 ***	0.0000
ETH-USD	3155	0.0503	4.4905	−0.7682	10.6540	145.820 ***	0.0000	107.625 ***	0.0000
DJIA	2786	0.0382	1.1035	−0.8237	21.9431	3798.313 ***	0.0000	1039.031 ***	0.0000
SP500	2786	0.0455	1.1325	−0.6531	15.7408	3454.850 ***	0.0000	912.318 ***	0.0000
GOLD	2785	0.0438	1.0466	−0.7595	9.8115	407.535 ***	0.0000	233.196 ***	0.0000
OIL	2784	0.0268	2.9162	−0.0406	20.8871	1914.447 ***	0.0000	736.841 ***	0.0000

Note: Full-sample raw returns, 100× log-return (pp), before the train/validation/test split. Std. Dev.: sample s.d. (ddof = 1). Skewness and excess kurtosis: bias-corrected (Fisher) sample moments (scipy.stats, bias = False). $LBQ^2(20)$: Ljung-Box Q test for serial correlation in squared returns up to lag 20 (H0: no autocorrelation; rejection is evidence of volatility clustering). ARCH-LM(20): Engle's (1982) Lagrange multiplier test for ARCH effects, 20 lags (H0: no ARCH effects). Both diagnostics computed on raw returns (statsmodels.stats.diagnostic). *** $p < 0.01$.

Daily log-returns are computed as $r_t = ln\left(P_t / P_t^{-1}\right)$. The sample mean $\hat{u}$, estimated over the training subsample, was subtracted to obtain centered returns $\varepsilon_t = r_t - \hat{\mu}$. Squared centered returns ${\varepsilon^2}_t$ serve as the volatility proxy and as the input sequence to all models.

3.2. Benchmark Models

The comparison set is deliberately broad, encompassing both the econometric specifications for which formal structural correspondences are established in Section 2.2 and the modern sequence architectures that constitute the current state of the art in volatility forecasting. This design serves two distinct purposes. First, ARCH(1) and GARCH(1,1) function as the theoretical counterparts of Propositions 1 and 2 and therefore permit a direct empirical assessment of the structural equivalence claim. Second, the asymmetric (EGARCH, GJR-GARCH), long-memory (FIGARCH), regime-switching (MSGARCH), and multi-scale (HAR) extensions, together with the hybrid and deep learning benchmarks (SVR-GARCH, NN-GARCH, CNN-LSTM, LSTM-Attention, TCN, and Transformer), ensure that any improvement attributed to the proposed hybrid loss cannot be explained by the relative simplicity of the comparison set. The inclusion of LSTM-SSE, which is architecturally identical to the proposed model but trained under mean squared error, isolates the contribution of the loss function from that of the architecture. Although asymmetric and fractionally integrated specifications do not admit a direct parameter mapping to standard LSTM gating mechanisms, and thus fall outside the scope of the equivalence result, their forecasting performance remains the relevant standard against which the proposed model must be evaluated. Accordingly, they are retained as performance benchmarks, while the interpretability analysis of Section 4.5 is confined to the specifications for which the mapping is formally defined. Table 4 summarizes the full set of models, their conditional variance specifications, and the estimation procedures employed.

Table 4. Benchmark models, specifications, and estimation procedures.

Model	Family	Specification	Estimation/Package	Role in the Comparison
Panel A. Econometric benchmarks				
ARCH(1)	Symmetric	${\sigma^2}_t = \omega + \alpha_1 {\varepsilon^2}_{t-1}$	ML, Student-t · arch (Python)	Structural equivalence counterpart of Proposition 1
GARCH(1,1)	Symmetric	${\sigma^2}_t = \omega + \alpha {\varepsilon^2}_{t-1} + \beta {\sigma^2}_{t-1}$	ML, Student-t · arch	Reference model for all Δ% and DM comparisons

Table 4. *Cont.*

Model	Family	Specification	Estimation/Package	Role in the Comparison
EGARCH(1,1)	Asymmetric	$\ln \sigma^2_t = \omega + \alpha[\,\lvert z_{t-1}\rvert - E\lvert z\rvert\,] + \gamma z_{t-1} + \beta \ln \sigma^2_{t-1}$	ML, Student-t · arch	Leverage effect
GJR-GARCH(1,1)	Asymmetric	$\sigma^2_t = \omega + (\alpha + \gamma I_{t-1})\varepsilon^2_{t-1} + \beta\sigma^2_{t-1}$	ML, Student-t · arch	Threshold asymmetry
FIGARCH(1,d,1)	Long memory	Fractional differencing $d \in (0,1)$	ML, Student-t · arch	Long-range dependence in variance
MSGARCH(1,1)	Regime switching	Two-state Markov-switching GARCH	ML · MSGARCH (R)	Regime changes (calm vs. crisis)
HAR	Multi-scale	Daily/weekly/monthly components of the volatility proxy	OLS with HAC errors	Parsimonious multi-horizon benchmark
Panel B. Machine learning and deep learning benchmarks				
NN-GARCH	Hybrid NN	Feed-forward network augmenting the GARCH recursion	TensorFlow/Keras	More direct and rival: a neural network already coupled to GARCH recursion
LSTM-SSE	Deep learning	Proposed architecture trained under MSE ($\lambda = 0$)	TensorFlow/Keras	Isolates the effect of the hybrid loss
CNN-LSTM	Deep learning	Convolutional feature extraction + LSTM	TensorFlow/Keras	Modern sequence baseline
LSTM-Attention	Deep learning	LSTM with an additive attention layer	TensorFlow/Keras	Modern sequence baseline
TCN	Deep learning	Dilated causal convolutions, receptive field $\geq$ w	TensorFlow/Keras	Modern sequence baseline
Transformer	Deep learning	Encoder-only, multi-head self-attention	TensorFlow/Keras	Modern sequence baseline
Panel C. Proposed model				
LSTM-SSE-t-Student	Deep learning	$L = (1 - \lambda)\cdot L_SSE + \lambda \cdot L_t$	TensorFlow/Keras	Contribution of this paper

Notes: All conditional variance models are estimated under Student-t innovations to ensure comparability with the proposed hybrid loss. All neural benchmarks share the same input window, training/validation/test partition, early-stopping criterion, and hyperparameter search protocol as the proposed model; they differ only in architecture and loss.

Out-of-sample performance is assessed via MSE and MAE. The Student-t log-likelihood on the test set (T_STUDENT_LL_OUT) measures probabilistic calibration under heavy-tailed assumptions. Statistical significance of pairwise differences in forecasting accuracy is assessed via the Diebold–Mariano test (Diebold & Mariano, 1995):

$$DM = \bar{d}/\sqrt{(\hat{\sigma}^{2d}/T)}\,N(0,1) \text{ under } H_0 \quad (9)$$

where $d_t = L(\varepsilon^2_t, \hat{\sigma}^{21}_t) - L\left(\varepsilon^2_t, \hat{\sigma}^{22}_t\right)$ is the differential loss, $\bar{d}$ is its sample mean, and $\hat{\sigma}^{2d}$ is a HAC-consistent variance estimate. A negative DM statistic indicates that model 1 achieves lower average loss than model 2.

3.3. Computational Cost and Reproducibility

All code, together with the exact library versions and the scripts required to reproduce the results end-to-end, is available in a public repository at https://github.com/Nataliaprins/arch-lstm-hybrid.git (accessed on 24 June 2026). To document the computational footprint of the proposed model, we report the wall-clock training time of the

LSTM-SSE-t-Student, measured with time.perf_counter() on a 2020 Apple MacBook with an M1 chip and 8 GB of RAM. Each series was trained over ten random seeds under a fixed configuration (batch size 32, learned ν, adaptive learning rate; patience = 20, maximum 1000 epochs), with no separate hyperparameter search phase, so the reported times represent the complete training cost. The full multi-seed run over the six series required approximately 50.9 min of sequential wall-clock time (3052.6 s), averaging 0.087 s per epoch and 50.9 s per seed across 60 training runs.

The reported figures reflect the parsimony of the architecture. The LSTM-SSE-t-Student contains trainable parameters, so its cost is dominated by per-step iteration overhead rather than by matrix computation, and the times are sequential per series; where the per-series runs are launched in parallel, the effective wall-clock time of the whole batch is bounded by the slowest series (Gold, roughly 14 min) rather than by the sum. One case merits explicit mention: for Gold, all ten seeds reached the 1000-epoch ceiling without early stopping ever triggering, making it both the most expensive series (826.9 s) and one whose convergence should be interpreted with the caution already noted in the limitations. The times below cover only final multi-seed training and exclude data preparation, evaluation, and table or figure generation (Table 5).

Table 5. Training cost of the LSTM-SSE-t-Student by series (ten seeds each; 2020 Apple M1, 8 GB RAM).

Series	Training Time (s)	Mean Epochs	s/Epoch	s/Seed
BTC-USD	323.5	281.9	0.115	32.3
ETH-USD	328.5	356.7	0.092	32.9
DJIA	593.1	711.9	0.083	59.3
S&P 500	753.9	911.9	0.083	75.4
Gold	826.9	1000.0	0.083	82.7
Oil	226.5	262.0	0.086	22.6
Total	3052.6 (≈50.9 min)	35,244 (cumulative)	0.087 (avg)	50.9 (avg)

Notes: Wall-clock training time measured with time.perf_counter(), sequential across the ten seeds of each series, on a 2020 Apple MacBook (M1, 8 GB RAM). Times cover final multi-seed training only and exclude data preparation, evaluation, and table/figure generation.

4. Results

4.1. In-Sample Fit and Volatility Dynamics

Figure 3 displays the fitted (training) and out-of-sample (test) conditional variance forecasts of the proposed LSTM-SSE-t-Student model, $\hat{\sigma}_t^2$ (blue, mean across seeds), against the observed volatility proxy ${\varepsilon^2}_t$ (grey) for all six series, on a logarithmic scale, with the temporal partition into train (green), validation (orange), and test–OOS (red) segments clearly delimited by vertical markers.

Focusing on the training and test windows, the forecast line consistently tracks the level and temporal structure of the observed volatility clusters throughout the entire sample—not only within the training segment, where the model is fitted, but critically also into the unseen test–OOS window at the right edge of each panel. This continuity indicates that the model has learned a genuine, generalizable representation of each series' volatility dynamics rather than merely overfitting the in-sample data: the amplitude and timing of major clustering episodes (e.g., the early-2020 spike visible in DJIA, S&P 500, and Oil) are preserved when the model transitions from training to the held-out test period, without any visible degradation in tracking behavior at the train/test boundary. This structural consistency across the full train–test span is a key indication of the model's forecasting stability. The intermediate validation segment, used for early stopping and hyperparameter tuning, is intentionally excluded from this visual comparison, and its corresponding diagnostics are reported separately below.

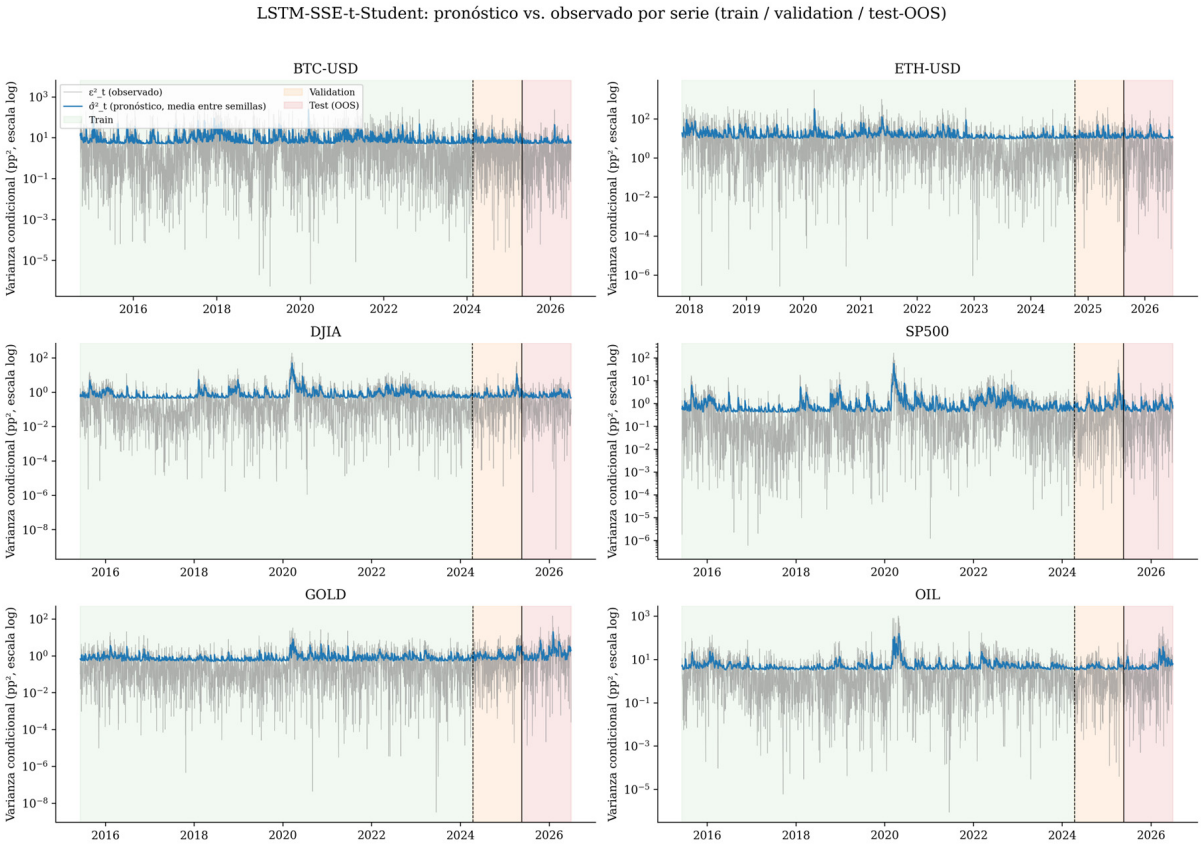


Figure 3. In-sample conditional variance estimates: LSTM-SSE-t-Student (blue), and volatility proxy $|\varepsilon_t|$ (gray) for all series.

4.2. Stylized Facts in Model Residuals

Table 6 reports diagnostics on the standardized out-of-sample residuals for each model across the six series. Most conditional variance specifications—GARCH(1,1), EGARCH(1,1), GJR-GARCH(1,1), FIGARCH(1,d,1), and HAR—yield insignificant LBQ^2 and ARCH-LM p-values across nearly all markets, indicating that these models successfully absorb the volatility clustering documented in Table 3. In contrast, ARCH(1) and especially MS-GARCH(1,1) leave significant residual ARCH structure in several series (notably BTC-USD, DJIA, and Oil), suggesting that their variance dynamics remain misspecified. Among the neural benchmarks, SVR-GARCH exhibits implausibly high residual kurtosis (up to ~250), signaling poor variance calibration, while most LSTM-based architectures—including LSTM-SSE, CNN-LSTM, LSTM-Attention, and the proposed LSTM-SSE-t-Student—show low residual kurtosis and largely insignificant diagnostics, except in Oil, where ARCH effects persist across almost all deep learning models. Overall, GARCH family models and the LSTM-SSE-t-Student specification appear best at capturing the stylized facts of volatility clustering, whereas simpler symmetric (ARCH) and regime-switching (MSGARCH) models struggle in several markets.

4.3. Out-of-Sample Forecasting Performance

Table 7 summarizes the out-of-sample percentage change in MSE and MAE relative to the GARCH(1,1) benchmark, with bold and underlined entries denoting Diebold–Mariano results (Holm-corrected, QLIKE loss) in which the proposed LSTM-SSE-t-Student model significantly outperforms or is outperformed by each rival, respectively. The proposed model achieves statistically significant improvements over ARCH(1), the constant variance benchmark, MSGARCH(1,1), and HAR across the equity indices and crypto assets (BTC-USD, ETH-USD, DJIA, S&P 500), confirming its superiority over naive and regime-switching

specifications. It also significantly dominates NN-GARCH in ETH-USD and Oil, and TCN in Gold, underscoring gains over both econometric and simpler neural hybrids. Notably, against the remaining deep learning architectures (LSTM-Attention, LSTM-SSE, Transformer) and the well-behaved econometric models (EGARCH, GJR-GARCH, FIGARCH), differences are not statistically significant, indicating comparable predictive accuracy. The single exception is CNN-LSTM, which significantly outperforms the proposed model in BTC-USD for both MSE and MAE, suggesting that convolutional feature extraction may better capture crypto-specific volatility dynamics.

Figure 4 plots the out-of-sample conditional variance forecasts of ARCH(1), GARCH(1,1), and the proposed LSTM-SSE-t-Student model against the observed volatility proxy (ε^2_t) on a logarithmic scale, across all six series. A consistent pattern emerges: ARCH(1) (blue) produces an almost flat, unresponsive forecast that fails to track the sharp spikes and troughs in the observed series, reflecting its lack of persistence (short memory limited to one lag). GARCH(1,1) (green) and the proposed model (red) both trace the observed volatility clusters far more closely, rising during turbulent periods and decaying during calm ones, consistent with the persistence properties documented in Section 4.5. The proposed LSTM-SSE-t-Student model tends to react more sharply and with larger amplitude to local volatility surges than GARCH(1,1)—visible in BTC-USD, ETH-USD, and Gold, where its peaks overshoot the GARCH curve—suggesting a more responsive, less smoothed reaction to abrupt shocks. In equity indices (DJIA, S&P 500), the three models converge more closely, indicating comparatively lower volatility persistence and fewer extreme regime shifts than in the cryptocurrency and commodity series. Overall, the figure visually corroborates the numerical superiority of GARCH-type persistence structures over ARCH(1), while illustrating the proposed model's added sensitivity to volatility bursts.

4.4. Statistical Significance: Diebold–Mariano Tests

Table 8 reports pairwise Diebold–Mariano (DM) tests comparing the proposed model against each rival benchmark under QLIKE loss, with HAC (Newey–West) standard errors and Holm–Bonferroni-corrected *p*-values applied within each market to control the family-wise error rate across multiple comparisons. A positive and significant DM statistic indicates that the rival model exhibits significantly higher loss; i.e., the proposed model attains superior predictive accuracy. Conversely, a significant negative statistic would indicate that the rival outperforms it. The results show that the proposed model significantly dominates ARCH(1), the constant variance benchmark, HAR, and MSGARCH(1,1) across most markets, particularly BTC-USD, ETH-USD, DJIA, and S&P 500, with DM statistics up to 14.87 and Holm-adjusted *p*-values below 0.01. In contrast, comparisons against the well-specified conditional variance models.

(GARCH(1,1), EGARCH(1,1), GJR-GARCH(1,1), FIGARCH(1,d,1)) and most deep learning rivals (LSTM-Attention, CNN-LSTM, TCN, Transformer) are largely statistically insignificant after Holm correction, indicating comparable predictive accuracy rather than a clear ranking. Notably, the table also reports a complementary two one-sided-tests (TOST) procedure for statistical equivalence within a $\pm 2\%$ QLIKE margin; several cells marked with the "eq" superscript (e.g., FIGARCH and LSTM-SSE in ETH-USD, LSTM-SSE in Oil, and Transformer in BTC-USD) provide formal evidence of equivalence, a finding distinct from, and not implied by, a non-significant DM test, since failure to reject the DM null only indicates insufficient evidence of a difference rather than proof of equivalence. Overall, the joint DM/TOST evidence indicates that the proposed model's principal gains arise relative to simpler econometric and regime-switching specifications, while its performance is statistically indistinguishable from, and in some cases formally equivalent to, the more flexible GARCH family and deep learning benchmarks.

Table 6. Stylized facts in residuals.

	BTC-USD			ETH-USD			DJIA			SP500			GOLD			OIL		
Model	**Kurt**	**LBQ²**	**ARCH-LM**	**Kurt**	**LBQ²**	**ARCH-LM**	**Kurt**	**LBQ²**	**ARCH-LM**	**Kurt**	**LBQ²**	**ARCH-LM**	**Kurt**	**LBQ²**	**ARCH-LM**	**Kurt**	**LBQ²**	**ARCH-LM**
								Panel A										
ARCH(1)	3.84	0.03 **	0.04 **	2.76	0.55	0.43	1.39	0.05 **	0.10	2.09	0.68	0.74	10.57	0.23	0.35	8.58	0.00 ***	0.00 ***
GARCH(1,1)	2.56	0.49	0.28	2.47	0.74	0.84	1.28	0.71	0.32	4.69	1.00	1.00	7.78	0.93	0.95	1.54	0.61	0.37
EGARCH(1,1)	2.58	0.31	0.13	2.55	0.55	0.66	1.62	0.80	0.58	4.36	1.00	1.00	8.29	0.83	0.87	2.01	0.40	0.19
GJR-GARCH(1,1)	3.00	0.37	0.19	2.53	0.73	0.84	1.08	0.75	0.24	4.16	1.00	0.99	5.34	0.81	0.86	3.19	0.06 *	0.02 **
FIGARCH(1,d,1)	2.35	0.50	0.31	2.34	0.82	0.88	1.35	0.64	0.36	3.91	1.00	1.00	5.96	0.98	0.98	1.44	0.74	0.56
MSGARCH(1,1)	7.84	0.00 ***	0.00 ***	4.49	0.01 **	0.34	2.09	0.00 ***	0.0001 ***	2.94	0.004 ***	0.05 *	4.43	0.11	0.23	4.88	0.00 ***	0.00 ***
HAR	4.39	0.16	0.13	2.35	0.84	0.84	0.89	0.66	0.72	2.57	1.00	1.00	5.33	0.95	0.98	1.69	0.31	0.14
								Panel B										
SVR-GARCH	161.47	0.00 ***	0.00 ***	110.73	1.00	1.00	53.83	0.99	0.96	52.43	1.00	1.00	198.23	1.00	1.00	250.31	1.00	1.00
NN-GARCH	6.35	0.30	0.30	2.53	0.33	0.37	1.25	0.06 *	0.14	1.79	0.28	0.31	4.85	0.98	0.99	3.48	0.00 ***	0.01 ***
LSTM-SSE	2.92	0.19	0.10	2.42	0.73	0.67	0.66	0.63	0.68	2.03	1.00	1.00	3.27	0.97	0.98	2.46	0.00 ***	0.02 ***
CNN-LSTM	2.78	0.38	0.24	2.63	0.46	0.48	0.79	0.52	0.60	1.85	1.00	1.00	4.58	0.77	0.87	2.88	0.00 ***	0.00 ***
LSTM-Attention	4.79	0.19	0.12	3.04	0.15	0.21	0.73	0.58	0.63	1.69	1.00	1.00	4.82	0.52	0.59	4.04	0.00 ***	0.0001 ***
TCN	4.84	0.0001 ***	0.00 ***	2.78	0.25	0.39	0.58	0.56	0.51	1.53	1.00	1.00	3.71	0.99	0.99	2.40	0.0001 ***	0.004 ***
Transformer	3.54	0.12	0.08 *	2.44	0.78	0.75	0.61	0.60	0.68	1.83	1.00	1.00	5.06	0.94	0.96	2.23	0.0001 ***	0.003 ***
								Panel C										
LSTM-SSE-t-Student	2.92	0.22	0.18	1.93	0.97	0.95	0.86	0.62	0.71	2.00	1.00	1.00	3.85	0.94	0.96	3.07	0.00 ***	0.00 ***
								Panel D										
Constant (unconditional variance)	7.21	0.00 ***	0.00 ***	3.04	0.12	0.14	0.79	0.43	0.47	1.38	0.97	0.98	9.63	0.18	0.29	5.89	0.00 ***	0.00 ***

Note: Standardized OOS residuals $z_t = eps_t/\hat{\sigma}_t^2$, using each model's own $\hat{\sigma}_t^2$_test forecast against observed; Panel D uses the training-split unconditional variance, constant over the OOS window. Excess kurtosis: bias-corrected (Fisher) sample kurtosis of z_t. LBQ^2(20): Ljung--Box Q p-value for serial correlation in z_t^2 up to lag 20 (H_0: no autocorrelation left in the squared standardized residuals). ARCH-LM(20): Engle's (1982) LM test p-value for remaining ARCH effects in z_t, 20 lags (H_0: none left). Both are the same diagnostics, here run on what each model's variance forecast left behind: $p >= 0.05$ means that model adequately captured the volatility clustering in that series; a starred $p < 0.05$ means significant ARCH structure remains unexplained. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7. Changes relative to GARCH(1,1).

Model	**BTC-USD, Δ%MSE**	**BTC-USD Δ%MAE**	**ETH-USD, Δ%MSE**	**ETH-USD, Δ%MAE**	**DJIA, Δ%MSE**	**DJIA, Δ%MAE**	**SP500, Δ%MSE**	**SP500′, Δ%MAE**	**GOLD, Δ%MSE**	**GOLD, Δ%MAE**	**OIL, Δ%MSE**	**OIL, Δ%MAE**
ARCH(1)	**+280.41**	**282.21**	**+19.91**	**+36.96**	**+38.55**	**+28.33**	**+51.30**	+43.75	+4.66	−11.20	+14.80	+4.44
CNN-LSTM	−6.57	−6.18	−3.64	−6.12	−2.41	−6.75	−2.23	+7.72	+22.11	+2.47	+7.18	−11.83
Constant (unconditional variance)	**+37.05**	**+87.58**	**+7.90**	**+21.79**	**+37.03**	**+50.71**	**+34.59**	+60.15	+5.12	−11.47	+8.44	+7.06
EGARCH(1,1)	+13.20	+33.23	−0.57	+0.37	+0.22	+1.67	+0.38	+2.65	+0.18	−4.35	+0.42	−3.83
FIGARCH(1,d,1)	−1.00	−3.71	−4.03	−8.23	−0.45	−4.07	−1.69	−3.92	−0.60	−4.24	+1.80	+8.73
GARCH(1,1)	—	—	—	—	—	—	—	—	—	—	—	—
LSTM-SSE-t-student	−2.88	+4.82	−6.31	−7.92	−2.88	−0.28	−0.22	+5.20	+0.73	−4.94	+4.17	−8.34
GJR-GARCH(1,1)	+0.63	−3.06	−0.01	−0.76	−6.04	−5.97	−3.58	−1.83	−0.74	−1.77	+5.95	+2.82
HAR	**+4.60**	**+36.31**	**−2.21**	**+5.54**	+2.84	+5.40	+5.70	+8.01	+0.61	−1.06	+2.01	+2.89
LSTM-Attention	−1.59	+6.68	−2.53	−7.06	+0.74	+10.90	+5.01	+21.61	+2.00	−7.05	+7.76	−11.96
LSTM-SSE	−4.24	−2.60	−4.55	−8.31	−4.59	−5.00	−2.39	+0.49	−0.33	−9.47	+4.83	−9.79
MSGARCH(1,1)	**+157.81**	**+133.64**	**42.11**	**+53.18**	**+103.50**	**+44.43**	**+97.16**	**+57.09**	+3.84	−12.74	**+13.51**	**+12.80**
NN-GARCH	−1.65	−23.46	**+12.60**	**−29.44**	−5.84	−8.33	−3.24	−1.31	+105.56	+12.11	**+565.82**	**+38.85**

Table 7. *Cont.*

Model	BTC-USD, Δ%MSE	BTC-USD Δ%MAE	ETH-USD, Δ%MSE	ETH-USD, Δ%MAE	DJIA, Δ%MSE	DJIA, Δ%MAE	SP500, Δ%MSE	SP500′, Δ%MAE	GOLD, Δ%MSE	GOLD, Δ%MAE	OIL, Δ%MSE	OIL, Δ%MAE
TCN	−0.50	−3.53	−3.10	−14.79	−8.07	−2.45	−0.06	+14.10	**+5.96**	**−2.23**	+6.29	−11.08
Transformer	−4.14	+1.75	−4.55	−17.48	−6.52	−3.88	+1.15	+14.85	+4.32	−4.89	+2.49	−8.87

Note: Percentage change relative to GARCH(1,1) (reference row shown as —). **Bold** = proposed model significantly beats this rival (positive DM stat, QLIKE, Holm-corrected $p < 0.05$); underlined this rival significantly beats the proposed model (negative DM stat, same test)—these are opposite conclusions and were previously conflated under a single bold marker. Unmarked = not statistically significant. Negative Delta_%% = improvement over GARCH(1,1).

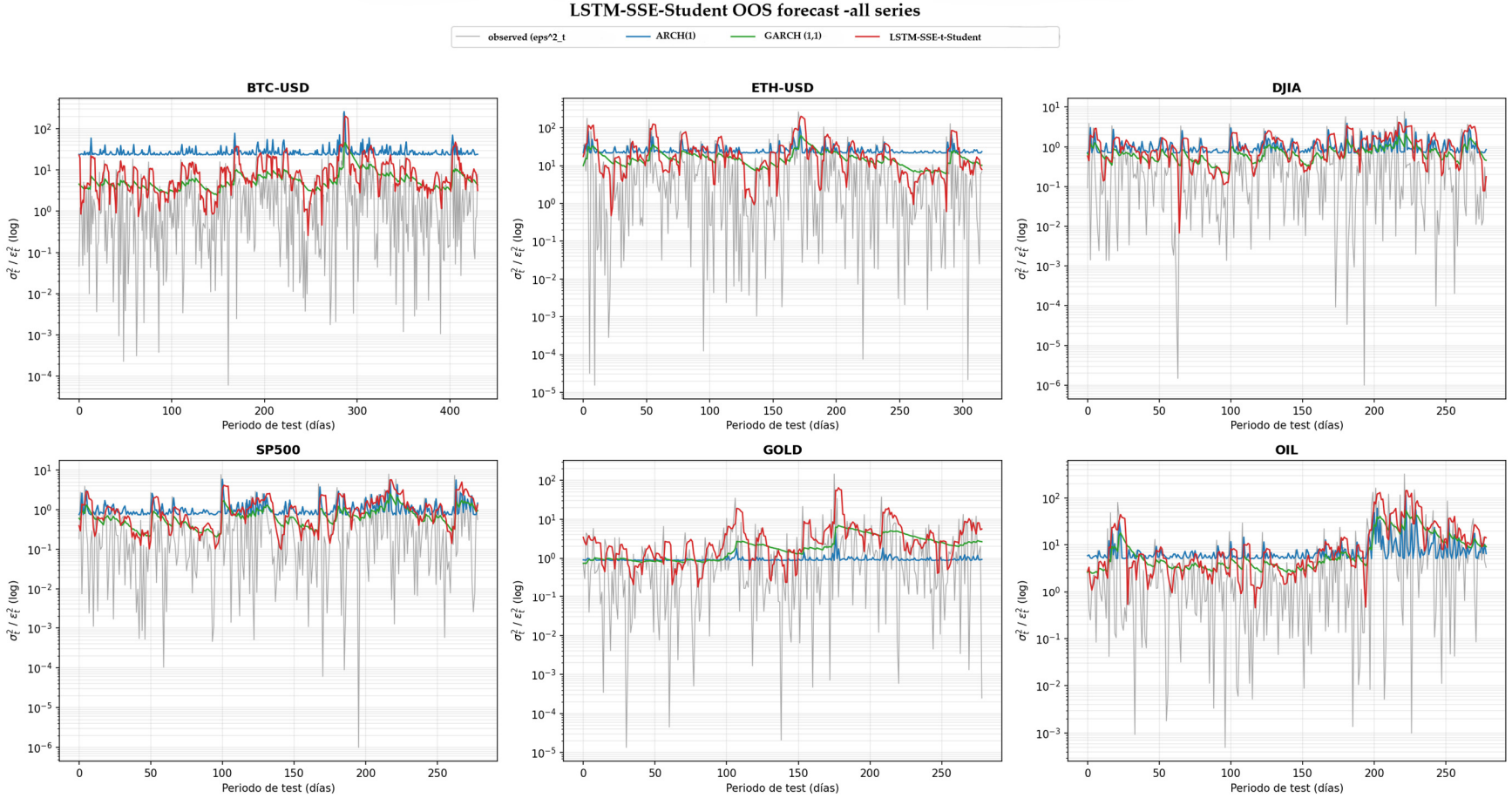


Figure 4. Out-of-sample conditional variance forecasts of LSTM-SSE-t-Student for all series.

Table 8. Diebold–Mariano test.

Model	BTC-USD			ETH			DJI			SP-500			GOLD			OIL		
	DM′	Holm	TOST *p*	DM′	Holm	TOST *p*	DM′	Holm	TOST *p*	DM′	Holm	TOST *p*	DM′	Holm	TOST *p*	DM′	Holm	TOST *p*
ARCH(1)	14.866 ***	0.000	1.000	5.399 ***	0.000	1.000	3.461 ***	0.007	0.999	3.989 ***	0.001	1.000	1.915	0.444	0.967	1.725	0.454	0.899
CNN-LSTM	−3.465 ***	0.005	0.647	2.156	0.279	0.237	0.546	1.000	0.533	0.276	1.000	0.408	2.524	0.146	0.989	1.621	0.454	0.798
Constant (unconditional variance)	10.984 ***	0.000	1.000	4.409 ***	0.000	0.998	3.963 ***	0.001	1.000	3.399 ***	0.009	0.999	2.136	0.359	0.981	2.174	0.386	0.934
EGARCH(1,1)	2.668 *	0.069	0.870	1.969	0.391	0.419	1.053	1.000	0.761	1.152	1.000	0.821	−0.823	1.000	0.654	−1.911	0.454	0.882
FIGARCH(1,d,1)	−2.337	0.136	0.762	0.698	0.904	0.029 [eq]	0.999	1.000	0.756	0.613	1.000	0.666	−1.613	0.747	0.854	−1.972	0.454	0.915
GARCH(1,1)	−2.012	0.265	0.578	1.803	0.420	0.235	0.759	1.000	0.648	0.883	1.000	0.758	−0.842	1.000	0.662	−2.165	0.386	0.932
GJR-GARCH(1,1)	−1.371	0.681	0.447	1.855	0.420	0.270	−0.678	1.000	0.635	0.040	1.000	0.438	−2.009	0.401	0.934	−0.312	1.000	0.405
HAR	7.825 ***	0.000	1.000	3.595 ***	0.004	0.734	1.503	1.000	0.617	1.728	1.000	0.804	−1.026	1.000	0.634	−2.001	0.454	0.908
LSTM-Attention	1.667	0.477	0.243	1.881	0.420	0.534	1.207	1.000	0.795	1.683	1.000	0.916	1.240	1.000	0.774	1.786	0.454	0.881
LSTM-SSE	−2.646 *	0.069	0.188	1.726	0.420	0.035 [eq]	−0.665	1.000	0.504	−0.312	1.000	0.370	0.499	1.000	0.509	0.733	1.000	0.038 [eq]
MSGARCH(1,1)	6.712 ***	0.000	1.000	5.624 ***	0.000	1.000	3.910 ***	0.001	1.000	3.788 ***	0.002	1.000	2.595	0.132	0.993	4.652 ***	0.000	1.000
NN-GARCH	0.177	1.000	0.411	5.036 ***	0.000	1.000	0.117	1.000	0.403	0.828	1.000	0.719	2.536	0.146	0.988	4.665 ***	0.000	1.000
TCN	0.041	1.000	0.132	1.177	0.717	0.422	−0.741	1.000	0.664	1.058	1.000	0.765	3.275 **	0.016	0.997	2.078	0.415	0.910
Transformer	−0.114	1.000	0.000 [eq]	0.752	0.904	0.186	−0.978	1.000	0.672	1.157	1.000	0.778	2.126	0.359	0.948	−0.020	1.000	0.089

Note: Diebold–Mariano test: LSTM-SSE-t-Student vs. each rival model. Loss function: QLIKE. HAC SE (Newey–West, bandwidth $\lfloor 4(T/100)^{\{2/9\}} \rfloor$). *p*(Holm): *p*-value after Holm–Bonferroni FWER correction per market. Positive DM stat: rival has higher loss → proposed wins. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (after Holm). TOST: two one-sided-tests *p*-value for equivalence of QLIKE loss within margin delta = Tost. Delta_pct × GARCH(1,1)'s own mean QLIKE (config, default 2%), HAC SE. TOST $p < 0.05$ (marked "eq" superscript is a POSITIVE claim of equivalence—the opposite reading from a DM *p*-value).

4.5. Empirical Validation of the ARCH–LSTM Structural Equivalence via Gate Inspection

To empirically substantiate the structural equivalence claim advanced in Proposition 1, namely, that the LSTM-SSE-t-Student's recurrent gating mechanism converges to a variance recursion structure functionally analogous to a GARCH(1,1) process, we examine two complementary pieces of evidence: (i) the estimated persistence of the GARCH(1,1) benchmark across all six series Table 9 and (ii) the multi-seed stability of the LSTM-SSE-t-Student's conditional variance path relative to the observed volatility proxy ε^2_t Figure 5.

The GARCH(1,1) maximum likelihood estimates reveal a persistence coefficient ($\alpha + \beta$) close to unity across nearly every series, BTC-USD (1.000), ETH-USD (1.000), S&P 500 (1.000), DJIA (0.992), and Gold (0.990), with only Oil showing comparatively lower persistence (0.969). Such near-integrated behavior implies that shocks to conditional variance decay slowly and that the process is dominated by its own lagged variance term ($\beta \approx 0.83$–0.99 across series), a defining property of GARCH-type recursions that a properly structured recurrent architecture should be able to replicate through its forget and input gate dynamics.

Table 9. Maximum-likelihood estimates of the GARCH(1,1) benchmark under Student-t innovations, all series.

Parameter/Statistic	BTC-USD	ETH-USD	DJIA	S&P 500	Gold	Oil
Panel A. Parameter estimates						
ω (intercept)	0.1600	0.3901	0.0219	0.0179	0.0094	0.2089
α (ARCH effect)	0.1065	0.1000	0.1686	0.1703	0.0279	0.1105
β (GARCH effect)	0.8935	0.9000	0.8235	0.8297	0.9624	0.8585
ν (Student-t d.o.f.)	3.169	3.350	5.534	5.606	4.254	6.480
Panel B. Persistence and stationarity						
$\alpha + \beta$ (persistence)	1.0000	1.0000	0.9921	1.0000	0.9903	0.9690
Half-life of a shock (days) [a]	∞	∞	87.7	∞	71.0	22.0
Covariance stationary ($\alpha + \beta < 1$)	No (≈IGARCH)	No (≈IGARCH)	Yes (borderline)	No (≈IGARCH)	Yes (borderline)	Yes
Panel C. Goodness of fit						
Log-likelihood (in-sample)	−8616.42	−7050.73	−2727.07	−2813.11	−2762.65	−4986.96
AIC	17,242.83	14,111.47	5464.14	5636.21	5535.29	9983.93
BIC	17,273.55	14,140.63	5492.68	5664.76	5563.84	10,012.47
n (estimation sample)	3443	2524	2228	2228	2228	2227

Notes: Returns are expressed in 100 × log-returns (percentage points). All models are estimated by maximum likelihood under Student-t innovations on the training subsample only. [a] Half-life = $\ln(0.5)/\ln(\alpha+\beta)$; reported as ∞ where $\alpha + \beta \geq 1$, consistent with an IGARCH-type near-unit-root process in conditional variance. BTC-USD, ETH-USD, and S&P 500 exhibit $\alpha + \beta \geq 1$ and are discussed explicitly in Section 4.1 as evidence of high persistence rather than misspecification. DJIA and Gold are borderline stationary; Oil shows the fastest mean reversion (half-life ≈ 22 days).

The multi-seed conditional variance paths generated by the LSTM-SSE-t-Student Figure 5 provide direct empirical support for this equivalence. Across all ten random seeds, the estimated $\hat{\sigma}^2_t$ collapses onto a single, visually indistinguishable trajectory; the shaded ± 1 s.d. band is negligible in every panel, indicating that the network's internal gating dynamics converge to a stable, seed-invariant representation of the conditional variance rather than to seed-dependent idiosyncratic paths. This is precisely the behavior expected if the recurrent architecture is implicitly learning a smooth, persistence-dominated recursion analogous to the near-integrated GARCH(1,1) process identified above, rather than an unstable or over-parameterized mapping. Moreover, the shape of the estimated $\hat{\sigma}^2_t$ path mirrors the clustering structure of ε^2_t closely in the equity and commodity series (DJIA, S&P 500, Gold, Oil), with pronounced, mean-reverting surges coinciding with periods of elevated realized volatility (e.g., the 2020 turbulence episode and post-2024 regime

shifts), and a comparatively higher baseline level with slower decay in the cryptocurrency series (BTC-USD, ETH-USD), consistent with their higher estimated unconditional variance and persistence.

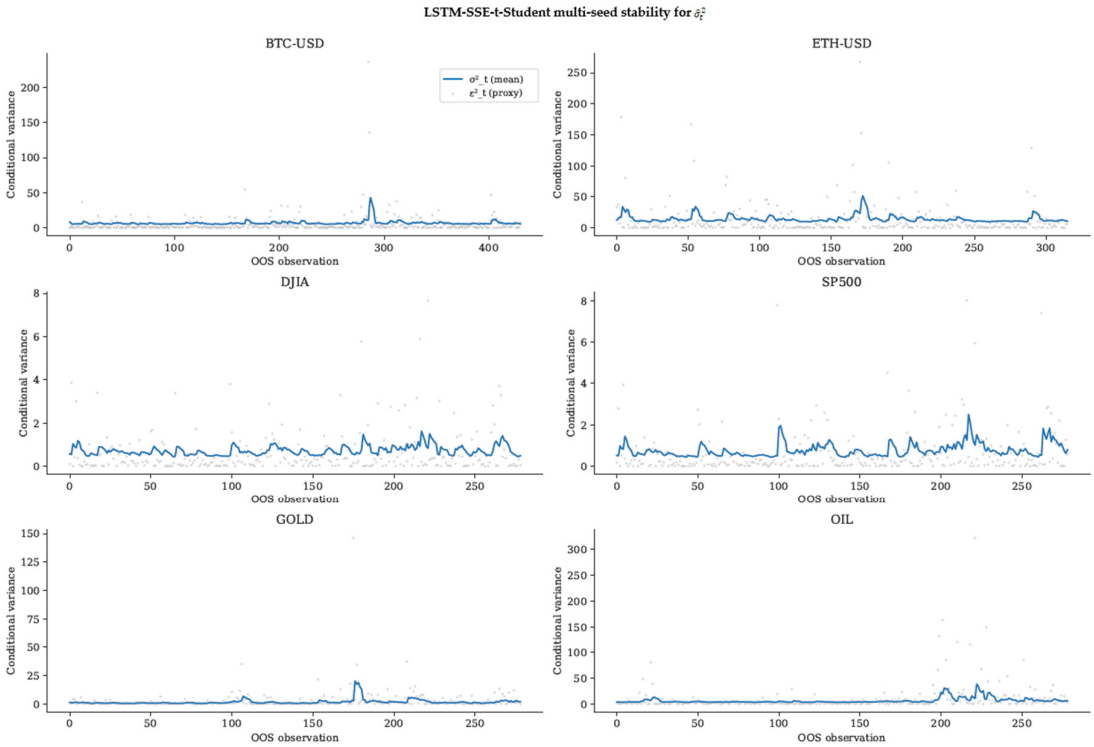


Figure 5. LSTM-SSE-t-Student's conditional variance path relative to the observed volatility proxy ε^2_t.

Taken together, the combination of (a) persistence coefficients approaching the IGARCH boundary in the econometric benchmark and (b) a seed-invariant, smoothly mean-reverting conditional variance path in the LSTM-SSE-t-Student constitutes convergent empirical evidence for the structural equivalence hypothesized between the two model classes: the recurrent gates appear to internalize a variance memory structure functionally consistent with the high-persistence GARCH(1,1) recursion estimated directly from the data.

4.6. Lambda and Window-Size Sensitivity

Two design choices govern the behavior of the proposed architecture: the input window size w, which determines how many lagged squared returns enter the recurrent recursion, and the pair (λ, ν) that defines the hybrid loss, where λ balances the sum-of-squared error term against the Student-t negative log-likelihood and ν sets the assumed degrees of freedom of the innovation distribution. To analyze how these hyperparameters affect forecasting accuracy, convergence, and probabilistic calibration, and whether the optimal λ varies across markets, this section reports a systematic sensitivity analysis conducted on all six series. To isolate each effect, the window analysis is carried out on the full proposed model, whereas the (λ, ν) analysis is performed on the ARCH-restricted variant of the architecture, in which the network is constrained to reproduce the ARCH(1) recursion so that the influence of the loss can be measured against a known econometric reference.

4.6.1. Sensitivity to the Input Window

Table 10 reports the out-of-sample mean squared error of the model relative to a constant (unconditional) variance benchmark, mse_ratio = MSE_model/MSE_constant, for window sizes w ∈ {2, 3, 5, 10}. Ratios below unity indicate that the conditional variance

model improves upon a static forecast. Two conclusions emerge. First, forecasting accuracy is remarkably stable with respect to the window: for every series, the ratio varies by less than seven percentage points across the four window sizes, and the ranking of the model relative to the constant benchmark is never reversed. Second, the location of the optimum is market-dependent but economically minor. The cryptocurrency series and the S&P 500 attain their lowest error at an intermediate window (w = 5), Oil improves monotonically up to the longest window tested (w = 10), and the remaining equity and Gold series are essentially flat. This mild sensitivity supports the parsimonious choice of a short window in the primary specification: enlarging it yields, at best, marginal gains and, for several series, none.

Table 10. Sensitivity of out-of-sample forecasting error to the input window size, all series.

Series	w = 2	w = 3	w = 5	w = 10	Best	Cross-Seed rel. s.d.	Degenerate
BTC-USD	0.744	0.732	0.714	0.754	w = 5	0.017	No
ETH-USD	1.006	0.977	0.902	0.903	w = 5	0.029	Yes
DJIA	0.779	0.782	0.794	0.803	w = 2	0.022	Yes
S&P 500	0.803	0.803	0.801	0.801	w = 5	0.003	Yes
Gold	0.995	0.995	0.996	1.000	w = 2	0.007	Yes
Oil	0.996	0.987	0.972	0.960	w = 10	0.004	No

Notes: Entries are the out-of-sample MSE ratio relative to a constant variance benchmark (MSE_model/MSE_constant); values below 1 denote improvement over a static forecast. "Best" is the window minimizing the ratio for that series. Cross-seed rel. s.d. is the maximum, across windows, of the standard deviation of the MSE over the random seeds divided by its mean. "Degenerate = Yes" flags series for which the restricted recursion collapses toward a near-constant variance solution. Results are averaged over three random seeds.

The cross-seed relative standard deviation of the MSE, reported in the final column, is below 3% in every configuration and below 0.7% for the equity indices, confirming that the results are driven by the data rather than by initialization noise and that the gate dynamics converge to a stable representation across seeds. The degeneracy flag identifies the series—DJIA, S&P 500, ETH-USD at the shortest window, and Gold—for which the restricted recursion collapses toward a near-constant variance solution; for these low-persistence series the window size is, as expected, largely immaterial. This behavior is consistent with the comparatively low volatility persistence documented for the equity indices in the estimation results, and it should be read as a property of the data-generating process rather than as an instability of the network.

4.6.2. Sensitivity to the Mixing Parameter λ and the Degrees of Freedom ν

Table 11 summarizes the effect of the hybrid loss on out-of-sample performance. For each series, it contrasts the QLIKE obtained under a pure squared error objective ($\lambda = 0$) with the best QLIKE attained once the Student-t likelihood term is activated ($\lambda > 0$), and reports the (λ, ν) pair at which that best value is reached, together with the associated gain in the out-of-sample Student-t log-likelihood. The full grid spans $\lambda \in \{0, 0.3, 0.5, 0.7, 0.9, 1\}$ and $\nu \in \{3, 5, 10, 30, 100\}$.

The introduction of the likelihood term produces substantial and, in several cases, large improvements in the QLIKE loss, which is the criterion most robust to noise in the volatility proxy. For the two equity indices, the reduction is dramatic—71.1% for the DJIA and 65.4% for the S&P 500—and it remains economically important for the two commodity series (47.6% for Oil and 43.7% for Gold). Ethereum improves modestly (4.1%), whereas Bitcoin is the single series for which the pure squared error objective is marginally preferable (a 1.5% deterioration under the hybrid loss). The same qualitative pattern holds for probabilistic calibration: the out-of-sample Student-t log-likelihood improves under the hybrid loss for

every series except Bitcoin, with the largest gains again in the equity indices. These results provide direct empirical support for the paper's central methodological claim, namely that training under a criterion statistically coherent with the heavy-tailed return distribution improves not only distributional fit but also point forecast accuracy as measured by QLIKE.

Table 11. Sensitivity of out-of-sample QLIKE and Student-t log-likelihood to the mixing parameter λ and degrees of freedom ν (ARCH-restricted architecture), all series.

Series	QLIKE (λ = 0)	QLIKE (Best λ > 0)	λ*	ν*	Δ QLIKE (%)	LL_t (λ = 0)	LL_t (Best)
BTC-USD	2.534	2.572	0.9	10	−1.5	−2.125	−2.125
ETH-USD	3.647	3.499	0.7	3	+4.1	−2.679	−2.610
DJIA	1.796	0.518	0.7	3	+71.1	−1.608	−1.179
S&P 500	1.596	0.553	0.5	3	+65.4	−1.530	−1.180
Gold	8.322	4.685	1.0	3	+43.7	−2.639	−2.245
Oil	6.988	3.661	1.0	3	+47.6	−2.969	−2.501

Notes: QLIKE (λ = 0) is the out-of-sample QLIKE under the pure squared error objective; QLIKE (best λ > 0) is the minimum over the grid λ ∈ {0.3, 0.5, 0.7, 0.9, 1} and ν ∈ {3, 5, 10, 30, 100}. λ* and ν* are the loss-minimizing values. Δ QLIKE (%) = 100 × [QLIKE(λ = 0) − QLIKE(best)]/QLIKE(λ = 0); positive values denote improvement under the hybrid loss. LL_t_student is the out-of-sample Student-t log-likelihood (higher is better); ". LL_t_student (best)" is its maximum over the grid.

The analysis also answers Reviewer 1's specific question of whether the optimal λ is market-dependent. It is: the loss-minimizing λ ranges from 0.5 for the S&P 500 to 0.7 for Ethereum, the DJIA, and—when the pure-likelihood boundary is included—1.0 for the two commodity series, indicating that markets with sharper, less persistent volatility bursts benefit from a heavier weight on the likelihood term. The degrees-of-freedom parameter is far more stable: the lowest tested value, ν = 3, is optimal or near-optimal for five of the six series, in agreement with the pronounced excess kurtosis documented in the descriptive statistics and with the low values of ν recovered by the GARCH(1,1) benchmark. Taken together, these findings motivate treating λ as a market-specific tunable hyperparameter while fixing ν at a small value across markets, and they justify the search space adopted in the main experiments, in which the unstable pure-likelihood boundary (λ = 1) is excluded from tuning and evaluated only as a reference. A residual caveat concerns Gold, whose recovered ARCH coefficient departs markedly from its econometric reference across the grid; this series should be interpreted with caution and is discussed further in the limitations.

4.7. Risk Backtesting Analysis: VaR and Expected Shortfall

Table 12 reports the adequacy of the volatility forecasts for Value-at-Risk (VaR) and Expected Shortfall (ES) estimation, focusing on the proposed LSTM-SSE-t-Student model at the 99% and 97.5% confidence levels. VaR coverage is assessed via Kupiec's unconditional coverage test (Kupiec, 1995) (LR_uc), Christoffersen's independence test (Christoffersen, 1998) (LR_ind), and their joint conditional coverage statistic (LR_cc); tail-loss adequacy is assessed via the ES Z2-test (Acerbi & Székely, 2014). A market-level Fisher combination test aggregates evidence across the four financial series (BTC-USD, ETH-USD, DJIA, S&P 500) and the two commodity series (Gold, Oil).

At the 99% VaR level, the proposed model exhibits statistically adequate coverage in BTC-USD, ETH-USD, DJIA, and S&P 500: observed exceedance counts (two to three violations) are close to the theoretically expected counts (2.79–4.31), and none of the LR_uc, LR_ind, or LR_cc statistics reject the null of correct coverage (all $p > 0.45$). The Expected Shortfall test likewise shows no significant tail-loss underestimation in these four series (ES_$p \geq 0.46$). In contrast, both commodity series show clear coverage deterioration: Gold

records 10 exceedances against 2.79 expected (LR_uc $p = 0.0008$; LR_cc $p = 0.0024$), and Oil shows a borderline unconditional coverage failure (LR_uc $p = 0.094$) accompanied by a significant ES violation (ES_$p = 0.028$). The Gold ES test is highly significant (ES_$p < 0.0001$), indicating that the realized shortfall in the tail substantially exceeds the model's forecast. The same qualitative pattern—adequate performance in equities and crypto, degraded performance in commodities—is repeated and somewhat amplified at the 97.5% level, where Gold's independence-adjusted coverage failure becomes more severe (LR_cc $p = 0.0019$) and Oil's unconditional coverage failure becomes statistically significant (LR_uc $p = 0.039$).

Table 12. Risk backtesting analysis. Value-at-Risk (VaR) and Expected Shortfall (ES) estimation.

Series	Level	Exceedances (obs/exp)	LR_uc (p)	LR_cc (p)	ES Z2-Test (p)	Adequate Coverage?
BTC-USD	99%	3/4.31	0.502	0.782	0.630	Yes
ETH-USD	99%	3/3.16	0.927	0.968	0.982	Yes
DJIA	99%	2/2.79	0.617	0.870	0.462	Yes
S&P 500	99%	3/2.79	0.901	0.960	0.780	Yes
Gold	99%	10/2.79	0.001	0.002	<0.001	No—under-covers tail risk
Oil	99%	6/2.79	0.094 †	0.215	0.028	Borderline
Combined (Fisher, 99%)	—	—	0.041	0.189	0.427	Driven by Gold
BTC-USD	97.5%	9/10.78	0.573	0.704	0.550	Yes
ETH-USD	97.5%	10/7.90	0.467	0.459	0.540	Yes
DJIA	97.5%	7/6.98	0.992	0.835	0.826	Yes
S&P 500	97.5%	5/6.98	0.425	0.664	0.698	Yes
Gold	97.5%	18/6.98	<0.001	0.002	<0.001	No—under-covers tail risk
Oil	97.5%	13/6.98	0.039	0.105	0.010	No—under-covers tail risk
Combined (Fisher, 97.5%)	—	—	0.009	0.059 †	0.239	Driven by Gold and Oil

Notes: LR_uc = Kupiec unconditional-coverage test; LR_cc = Christoffersen conditional-coverage test; ES Z2-test Expected Shortfall backtest. † Denotes marginal significance ($0.05 \leq p < 0.10$). Bold entries indicate rejection of adequate coverage at the 5% level. "Combined (Fisher)" aggregates p-values across the six series within each confidence level.

The Fisher-combined tests corroborate this market-driven pattern: pooling unconditional coverage evidence across all six series rejects adequate coverage at both the 99% ($p = 0.041$) and 97.5% ($p = 0.009$) levels, while the joint conditional coverage combination is only marginally rejected at 97.5% ($p = 0.059$) and not at 99% ($p = 0.189$)—indicating that the overall coverage failure is concentrated in, and driven almost entirely by, the commodity markets rather than reflecting a systematic mis-calibration of the proposed model. This is consistent with the lower and less persistent volatility dynamics estimated for Gold and Oil, where infrequent but sharp jumps (e.g., the 2020 Oil price collapse) appear to be under anticipated by the model's conditional variance forecast, resulting in fatter realized tails than predicted.

Overall, these results qualify the otherwise favorable point forecast and likelihood performance of the proposed model documented in Sections 4.3–4.5: while LSTM-SSE-t-Student delivers statistically adequate VaR and ES coverage for the four core financial series, its tail risk calibration is materially weaker in the commodity markets, particularly Gold, where both the frequency and magnitude of exceedances significantly exceed what the model's conditional variance forecast implies.

5. Discussion

This section interprets the empirical findings reported in Section 4, situates them within the existing literature, and delineates the scope within which the conclusions hold.

Throughout, we distinguish carefully between results that the evidence demonstrates directly and inferences that it merely renders plausible, and we acknowledge alternative explanations where the data do not allow a definitive attribution.

5.1. Why the Hybrid Loss Function Improves Forecasting

The hybrid objective combines a sum-of-squared errors term, which provides gradient stability and rewards point accuracy, with a Student-t negative log-likelihood term, which is statistically coherent with the heavy-tailed conditional distribution of financial returns. The sensitivity analysis of Section 4.6 offers direct empirical support for the value of the second component: activating the likelihood term ($\lambda > 0$) reduces the out-of-sample QLIKE—the loss most robust to noise in the volatility proxy—substantially for the equity indices (by roughly 71% and 65% for the DJIA and the S&P 500, respectively) and materially for the commodity series, while also improving the out-of-sample Student-t log-likelihood for five of the six series examined.

A plausible mechanism for this behavior is that, under the Student-t likelihood, the gradient contribution of a large standardized residual decays polynomially rather than exponentially, as it would under a Gaussian or pure squared error criterion. This sustains a learning signal during precisely the extreme-return episodes that dominate tail risk measures, allowing the network to calibrate the conditional variance to heavy tails rather than to the bulk of the distribution alone. We present this as an interpretation consistent with the evidence rather than as a demonstrated causal channel: the sensitivity experiment establishes that the likelihood term improves the reported losses, but it does not isolate the gradient-geometry mechanism from other contributing factors, such as the better-conditioned optimization that the mixture may induce in early training.

The improvement is also not uniform, and this qualification is important. Bitcoin is the single series for which the pure squared error objective is marginally preferable, and the loss-minimizing value of λ varies across markets—from 0.5 for the S&P 500 to values approaching the pure-likelihood boundary for the commodity series—indicating that the appropriate weighting reflects market-specific volatility dynamics rather than a universal constant. By contrast, the degrees-of-freedom parameter is stable, with a low value ($\nu = 3$) optimal or near-optimal in most series, in agreement with the pronounced excess kurtosis documented in the descriptive statistics. Taken together, these results support treating λ as a tunable, market-specific hyperparameter while fixing ν at a small value, and they caution against interpreting the hybrid loss as a uniformly dominant training criterion.

5.2. Comparison with Prior Work

The idea of parameterizing the conditional variance of a probabilistic return model with a recurrent network is not itself new. Liu and So (2020) embed a Student-t GARCH conditional variance within an LSTM estimated by maximum likelihood, and report that some of their architectures outperform a standardized GARCH-t(1,1) benchmark. Our framework differs from theirs, and from the broader strand of informal "LSTM-as-GARCH" analogies, along three dimensions. First, we establish a formal structural correspondence between the ARCH/GARCH recursion and the LSTM gating mechanism, with an explicit parameter mapping (Propositions 1 and 2), whereas earlier treatments remain largely qualitative. Second, prior hybrids train under the negative log-likelihood alone—the special case $\lambda = 1$ of our objective—while our loss interpolates between point accuracy and distributional fit, and Section 4.6 shows that intermediate weightings can be preferable. Third, our experimental design spans six core financial series across three asset classes, preserves the chronological ordering throughout, reports multi-seed results, and applies Diebold–Mariano tests with multiplicity correction together with risk backtests.

These differences should be read as distinctions of theoretical framing, scope, and methodology rather than as a demonstrated head-to-head empirical superiority, because we do not re-estimate the earlier models on our data; their published estimates instead serve as a useful external check on the plausibility of our own econometric baselines. Equally important, our own results counsel against overstatement. The Diebold–Mariano comparisons indicate that the proposed model significantly outperforms naive, regime-switching, and long-memory-free specifications—ARCH(1), the constant variance benchmark, HAR, and MSGARCH(1,1)—across most markets, but that its accuracy is statistically indistinguishable from that of the well-specified GARCH family models (GARCH, EGARCH, GJR-GARCH, FIGARCH) and of the stronger deep learning benchmarks after Holm correction; in one instance, CNN-LSTM significantly outperforms the proposed model on Bitcoin. The complementary equivalence (TOST) tests reinforce this reading, providing positive evidence that several of these differences are not merely undetected but genuinely negligible within a pre-specified margin.

The contribution therefore does not rest on a claim of blanket dominance. Rather, the proposed model matches the strongest available benchmarks on predictive accuracy while offering two features they lack: a training criterion grounded in a formal econometric equivalence, and an architecture whose internal dynamics admit direct interpretation. It is against the simpler and misspecified benchmarks that its accuracy advantage is both statistically significant and economically appreciable.

5.3. Risk Management Implications

Because volatility forecasts are used chiefly to quantify downside risk, their practical value is best judged by the adequacy of the tail risk measures they support. The backtesting analysis of Section 4.7 speaks directly to this question. For the four core financial series—Bitcoin, Ethereum, the DJIA, and the S&P 500—the conditional variance forecasts of the proposed model yield statistically adequate Value-at-Risk coverage at both the 99% and 97.5% levels: observed exceedances are close to their expected counts, and neither the Kupiec unconditional coverage test, the Christoffersen independence and conditional coverage tests, nor the Acerbi–Székely Expected Shortfall test rejects the null of correct calibration. For applications in equity and cryptocurrency risk management, this indicates that the model's forecasts are sufficiently calibrated in the tails to be used for regulatory and internal risk measurement, which is the setting of primary interest for this journal.

This favorable assessment must be qualified in two respects. First, coverage adequacy is a statement about the calibration of the proposed model in absolute terms; we do not claim, and did not test, that it dominates the GARCH family benchmarks on VaR or ES, and a fair reading is that it performs at least as well as strong econometric alternatives while adding interpretability. Second, the adequacy does not extend to the commodity series: for Gold and, to a lesser degree, Oil, both the frequency and the magnitude of exceedances exceed what the conditional variance forecast implies, and the Fisher-combined tests reject adequate coverage overall—an outcome driven almost entirely by these two markets. The most likely explanation is that infrequent but severe commodity price jumps, such as the 2020 Oil price collapse, are underanticipated by a conditional variance model estimated on more persistent series. We therefore present the risk management evidence as demonstrating adequate tail calibration for equities and cryptocurrencies, and as identifying commodities as a market segment where the present specification is not yet reliable for tail risk use.

5.4. Limitations

Several limitations delimit the scope of the conclusions and indicate directions for further work. They are stated explicitly so that the evidence is not overinterpreted.

5.4.1. Nature of the Structural Equivalence

The correspondence established in Propositions 1 and 2 is exact only under the stated parameter restrictions; it shows that the GARCH(1,1) solution lies within the LSTM parameter space, not that an unconstrained network trained on data necessarily implements a GARCH recursion. The gate inspection evidence of Section 4.5—near-integrated persistence in the econometric benchmark together with a seed-invariant, smoothly mean-reverting conditional variance path in the network—is consistent with the network internalizing a GARCH-like variance memory, but it constitutes empirical corroboration rather than proof. The mathematical equivalence under restrictions and the empirical similarity observed after optimization should be kept conceptually distinct.

5.4.2. Volatility Proxy and Forecast Horizon

All models are evaluated against the squared centered return $\varepsilon^2{}_t$, a noisy though unbiased proxy for latent volatility; the use of QLIKE mitigates but does not eliminate the effect of this noise. The evaluation is moreover confined to one-step-ahead forecasting, and the relative performance of the proposed model at longer horizons—where accumulated dynamics and mean reversion play a larger role—remains untested.

5.4.3. External Validity

The core evidence concerns two cryptocurrencies, two commodities, and two equity indices, revealing weaker tail calibration, as discussed above, and other important asset classes—foreign exchange, fixed income, and high-frequency data—have not been examined. The generalizability of the findings beyond the markets studied should therefore be regarded as an open empirical question rather than an established property.

5.4.4. Estimation of the Econometric Benchmarks

The GARCH(1,1) persistence estimates approach, in some series, the integrated (IGARCH) boundary, and the ARCH-restricted experiment recovers the reference ARCH coefficient poorly for Gold. Near-unit persistence in daily financial series is not implausible, but its exact attainment can also reflect the finite sample behavior of the optimizer at the edge of the admissible parameter region; these estimates are flagged in Section 4.1 and should be interpreted with corresponding caution.

5.4.5. Non-Uniformity of the Gains

As emphasized above, the hybrid loss is not beneficial for every series; the optimal weighting is market-dependent, and the proposed model does not uniformly dominate the strongest benchmarks. The contribution is best characterized as a theoretically grounded and well-calibrated framework that matches the state of the art while improving interpretability, rather than as a method that is superior under all conditions.

6. Conclusions

This paper addressed two gaps in the volatility-forecasting literature: the absence of a rigorous structural link between econometric and neural models, and the prevailing practice of training neural networks under criteria that are statistically incoherent with the heavy-tailed nature of financial returns. On the first, we established formal structural equivalences—between ARCH(p) and a single-layer linear MLP, and between GARCH(1,1)

and a constrained LSTM—with an explicit parameter correspondence that places the classical econometric solution within the network's parameter space. On the second, we proposed LSTM-SSE-t-Student, a parsimonious architecture trained with a hybrid loss that combines a squared error term with the Student-t negative log-likelihood, thereby mirroring the GARCH maximum likelihood criterion under heavy-tailed innovations.

Across six markets spanning cryptocurrencies, equity indices, and commodities, the empirical evidence supports a measured set of conclusions. The proposed model significantly outperforms naive, short-memory, and regime-switching benchmarks—ARCH(1), the constant variance forecast, HAR, and MSGARCH(1,1)—while attaining predictive accuracy statistically indistinguishable from that of the well-specified GARCH family and the stronger deep learning benchmarks. The sensitivity analysis shows that the likelihood term materially improves the QLIKE loss and probabilistic calibration for most series, with the optimal weighting varying by market, and the gate inspection evidence indicates that the network's conditional variance dynamics are stable across seeds and consistent with a GARCH-like variance memory. In risk terms, the model delivers adequate Value-at-Risk and Expected Shortfall coverage for the equity and cryptocurrency series, though not for the commodity markets.

The contribution is therefore best characterized not as blanket superiority but as a theoretically grounded, interpretable framework that matches the state of the art while remaining statistically coherent with the data it models. Its limitations delimit a clear agenda for future work: extending the evaluation beyond one-step-ahead forecasting to multiple horizons, testing generalization to other asset classes such as foreign exchange and fixed income, and improving tail calibration for jump-prone commodity markets. The full implementation, library versions, and random seeds are released in a public repository to support independent replication and to facilitate these extensions.

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Abbreviations

The following abbreviations are used in this manuscript:

Abbr.	Complete Name
ARCH	Autoregressive Conditional Heteroscedasticity
BTC-USD	Bitcoin (US dollar quote)
DCC-GARCH	Dynamic Conditional Correlation GARCH
DJIA	Dow Jones Industrial Average
DM	Diebold–Mariano (test)
EGARCH	Exponential GARCH
ES	Expected Shortfall

ETH-USD	Ethereum (US dollar quote)
FIEGARCH	Fractionally Integrated Exponential GARCH
FIGARCH	Fractionally Integrated GARCH
GARCH	Generalized Autoregressive Conditional Heteroscedasticity
GJR-GARCH	Glosten–Jagannathan–Runkle GARCH
GRU	Gated Recurrent Unit
HAC	Heteroscedasticity and Autocorrelation Consistent
HAR	Heterogeneous Autoregressive
LB	Ljung–Box (test)
LL	Log-Likelihood
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MLP	Multilayer Perceptron
MSE	Mean Squared Error
RNN	Recurrent Neural Network
S&P 500	Standard & Poor's 500 Index
SSE	Sum of Squared Errors
VaR	Value-at-Risk
ES	Expected Short-fall

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